

Since the product of our project will be shown through a jupyter notebook, our plan will not consist of descriptions for each method, rather it will show the code we plan to use for a few code blocks at a time.

First code blocks: Imports

We will need to import a few libraries for our code to work. These libraries will be pandas, matplotlib, and seaborn for reading in the dataset/visualization and ski-kit learn for the models.

Second code blocks: Loading in the dataset/filtering bad values

Using pandas', we can read in our .csv file and use that dataset throughout the rest of our code. Looking through the dataset, we found a few duplicate rows as well. Using pandas, we can drop these duplicated rows so we are only using unique data points for our graphs.

Third code blocks: Heatmap

Seaborn has a heatmap command that prints out a graph to show how correlated each variable is. It can give the user an idea of which variable can directly impact another variable.

Fourth code blocks: Count Plots

Seaborn can also produce count plots, which in our case, can just add the total number of every variable we have that isn't the output variable (more or less likely to have heart disease). With this count plot, we can graph the prevalence of heart attack by each variable against the output variable, which will allow us and the user to draw direct comparisons.

Fifth code blocks: Hist plots

Similar to the count plots, seaborn can also produce histogram plots, which can also show the prevalence of heart attack by some of the variables we have. However, some of the variables will work better than others in this scenario, such as age, heart rate, and cholesterol levels. As with all of our graphs/plots, matplotlib gives us the ability to add titles, change the size, give the graphs placement, etc.

Sixth code blocks: Joint plots

The last seaborn plots that we plan to show are their joint plots. With their joint plots, seaborn will take two overlapping histograms of two of the qualitative variables and find where they overlap the most. For example, if we do a joint plot with the cholesterol and maximum heart rate levels, we can find a max. heart rate level that overlaps the most with a cholesterol level.

Seventh code blocks: Getting the data ready to split into groups

Using the 'train_test_split' parts of ski-kit learn, we can create training and testing variables that can amount to 80% and 20% of the data. We can have x and y values where the x values can be all of the data comprising the variables and the y value can comprise the output variable only.

One thing we have to keep in mind is that all of the data has to be turned into numerical data if they weren't already.

Eighth code blocks: First two models (1st CV credit)

Using the text and markdown features of Jupyter, we can write a brief description of each model and how they work before we conduct that part of the code.

*For the notebook overall, we can also add headers and any important information that we think of along the way for the user to see as they progress through our project.

Anyway, we will apply two models first and output their accuracy. These two models are going to be a Logistic Regression and a KNeighborsClassifier.

Ninth code blocks: 3rd and 4th models (2nd CV credit)

Continuing on with the models, we will apply the next two models, which are going to be a SVC and a DecisionTreeClassifier and show their accuracy.

Tenth code blocks: 5th and 6th models (3rd CV credit)

Lastly, we will apply a RandomForestClassifier and a GradientBoostingClassifier. We are confident that these 6 models are good ones that we can apply with the type of data that we have. As a last step, we can graph a bar plot with all of the accuracies next to each other so the user can see which model did the best. Lastly, seaborn offers a distplot in which we can likely plot the predicted values of our best model and the actual values of the data as a curve.